

Where to Go Next: Site Selection for External Validity via Sensitivity Analysis

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Abstract

A set of randomized experiments identifies the treatment effect at a new site only in part. [Boyarsky et al. \[2026\]](#) develop a sensitivity framework for this problem. Under a site-level fixed-effects response model, the observed sites determine the fixed effects only inside the span of their centered context profiles, and the framework reports the smallest and largest target effects consistent with the source experiments and with restrictions the analyst states. This paper studies the natural next question. Given several candidate sites whose contexts are known but whose experiments have not been run, which site would most strengthen the external validity claim of the multi-site study. We show that the benefit a new site provides depends on the affine hull of the site profiles. The target effect is point identified exactly when the target profile lies in the affine hull of the observed profiles, and a candidate site tightens the identified set only if its profile lies outside that hull. We show how we can forecast the benefit that a new site provides through the vector of site-level data. We apply the method to the multi-site credit expansion data of [Meager \[2019\]](#).

1 Introduction

Randomized experiments are often run in sites to establish evidence of an intervention applied in another site. When site-level context changes the treatment response, a pool of internally valid experiments identifies the effect at a new site only in part. [Boyarsky et al. \[2026\]](#) formalize this problem. Site context enters each potential-outcome surface through linear fixed effects, the observed sites determine those fixed effects only inside the span of their centered context profiles, and the target effect is reported as an interval: the smallest and largest values consistent with the source experiments and with restrictions the analyst is willing to state. A wide interval means that the data and the stated restrictions do not settle the target conclusion. An infinite interval means that some direction of extrapolation is not controlled at all.

An analyst facing a wide interval has two options. She can impose stronger restrictions, which tightens the set at the cost of assumptions that the data cannot check. She can also collect more evidence by running an experiment in a new site. The companion paper studies the first option. This paper studies the second. The question is concrete because site context is observable before any experiment is run. A funder choosing among candidate countries already knows each country's interest rates, loan sizes, and program rules. What she does not know is what an experiment there would find. We ask which candidate site, if its experiment were run, would most strengthen the external validity of the pooled evidence, and how much of that answer is available before the experiment.

We use the microcredit experiments harmonized by [Meager \[2019\]](#) as a running example. Seven countries ran nominally similar expansions of credit access under very different lending terms. The

companion paper’s case study holds out one country at a time and asks what the other six imply for it. Its leave-one-out exercise shows the reverse of our question. Dropping any one of Bosnia’s six source sites leaves Bosnia’s interval unsigned. That finding also says that the sixth experiment was worth running. We ask whether an analyst who had only five source sites could have known this in advance, from lending terms alone, and could have chosen the right country to add.

The paper has three main findings. Each follows from the identification geometry of the companion paper.

First, site selection is a question about the affine hull of the site profiles. In the companion framework the observed sites identify the fixed effects only inside the span of their centered profiles, and we show that the target effect is point identified exactly when the target profile lies in the affine hull of the observed profiles (Section 4). A candidate site changes what is identified only if its profile lies outside that hull. Whether it does is checkable at design time, before any data are collected. How far outside it lies does not matter for identification, because any experiment along the new direction pins down the same coordinate of the fixed effects. Distance affects precision but not the identified set.

Second, the value of a candidate experiment is computable in advance up to one scalar per treatment arm (Section 5). Under the maintained model, adding site k replaces the feasible set $\mathcal{F}$ by $\mathcal{F}$ intersected with two equality constraints, $u_k^\top \beta_a = y_{a,k}$ for $a \in \{0, 1\}$, where the direction u_k is known from the candidate’s context and the scalars $y_{a,k}$ are what the experiment reveals. The whole map from those scalars to the post-addition bounds can be solved before the experiment. Forecasting the scalars from the existing pool gives a predicted interval, and the dual variable of the added constraint gives the effect of a forecast error on the bounds. This is the design rule we apply in the simulations and the case study.

Third, under the maintained model a new site can never hurt. Every width weakly shrinks, at every target (Section 6). This may seem obvious, but it fails once the model is wrong. A site whose outcomes conflict with the shared response model moves the pooled moment conditions to a compromise between the sites. The identified slice then translates, and the width can grow or shrink, or the program can become infeasible. We give a two-site example where one conflicting addition doubles the width. The effect of the conflict on correctness is more important than its effect on the width. The translated set stops covering the true effect exactly at the targets the pool point identifies. The screens we propose, a rank test on profiles and a compatibility check on outcomes, separate the additions that refine the set from the additions that corrupt it.

We verify each claim with the estimator the companion paper recommends, not only with population calculations. The simulations in Section 8 generate multi-site data from the fixed-effects model, estimate the bounds with the cross-fitted rank consistent estimator, and show that the estimated selection recovers the population-optimal candidate in every replicate, that the design-time predictions match the realized widths, and that the conflict mechanism behaves as the theory says. The case study in Section 9 runs the same exercise on the microcredit data. For each five-country source pool, the rule scores the missing country from its lending terms, and the realized experiment confirms the prediction.

Related work. The site-selection question is the forward version of the leave-one-out exercise in the case study of [Boyarsky et al. \[2026\]](#), which asks which observed sites a reported bound depends on. We ask which unobserved site a future bound would benefit from, and we use the same three categories of what a site can do: expand identification, add precision, or break feasibility. The problem is also related to classical optimal experimental design [[Atkinson et al., 2007](#), [Pukelsheim, 2006](#)]. There the design variable is an allocation of observations inside one population and the loss

is a variance. Here the design variable is a site-level context profile and the loss is the width of a partial-identification interval, which variance reduction alone cannot shrink. Submodular coverage arguments for sensor placement and subset selection [Nemhauser et al., 1978] appear when several sites are added at once (Section 10). Distributionally robust formulations of worst-case effects [Jeong and Namkoong, 2020] and structural transportability conditions [Bareinboim and Pearl, 2016] address what can be learned from fixed data. Our contribution is to treat the data that do not yet exist as the design object.

2 Setting

Throughout this paper we use the setting and sensitivity framework of Boyarsky et al. [2026]. This section restates this setting.

The source sample pools S sites. An observation is $O = (Y, A, X, C)$, where $A \in \{0, 1\}$ is treatment, X contains unit-level covariates, and $C \in \mathbb{R}^p$ is the site-level context vector, constant within a site. The basis of C is fixed before the analysis and may contain raw site characteristics and their transformations. Site s carries the profile $c_s \in \mathbb{R}^p$. The source law P mixes the sites, and treatment is randomized within each site with known probability $e_a(x, c) = \mathbb{P}_P(A = a \mid X = x, C = c)$ bounded away from zero. A target site is defined by a pair (Q_T, c_T) : a covariate law we can sample and a context profile $c_T \in \mathbb{R}^p$. Treatment and outcome data exist only for the source sites.

The framework rests on a site-level fixed-effects response model.

Assumption 1 (Site-level fixed effects, Boyarsky et al., 2026). *For $a \in \{0, 1\}$,*

$$\mathbb{E}_P[Y(a) \mid X, C] = g_a(X) + C^\top \beta_a, \quad (2.1)$$

where g_a is unrestricted and $\beta_a \in \mathbb{R}^p$ is fixed. Treatment is randomized within each source site, consistency holds, the same functions and coefficients govern the target site, and all second moments used below are finite.

Write $R_a = \mathbf{1}\{A = a\}Y/e_a(X, C)$ for the inverse-probability pseudo-outcome, $m_C(x) = \mathbb{E}_P[C \mid X = x]$, and $\tilde{C} = C - m_C(X)$. Averaging (2.1) over the target population expresses the target average treatment effect as

$$\begin{aligned} \psi_T &= b_T + \ell_T^\top (\beta_1 - \beta_0), & b_T &= \mathbb{E}_{X \sim Q_T} [\mathbb{E}_P[R_1 - R_0 \mid X]], \\ \ell_T &= c_T - \mathbb{E}_{X \sim Q_T} [m_C(X)]. \end{aligned} \quad (2.2)$$

The scalar b_T is the covariate-transported part and is identified under overlap in X . The loading ℓ_T multiplies the fixed-effect contrast, which is the only unknown. Each β_a satisfies the normal equations of a partially linear regression [Robinson, 1988],

$$\Sigma_C \beta_a = d_a, \quad \Sigma_C = \mathbb{E}_P[\text{Cov}(C \mid X)], \quad d_a = \mathbb{E}_P[\text{Cov}(C, R_a \mid X)]. \quad (2.3)$$

Because C takes only S values, every residual $\tilde{C}$ lies in the span of the centered site profiles,

$$\mathcal{V}_C = \text{span}\{c_s - c_1 : s = 2, \dots, S\}, \quad r = \dim \mathcal{V}_C \leq S - 1, \quad (2.4)$$

so $\text{range}(\Sigma_C) \subseteq \mathcal{V}_C$ and $\text{rank}(\Sigma_C) \leq S - 1$. When $p > S - 1$ the normal equations leave fixed-effect directions undetermined, and ψ_T is point identified if and only if $\ell_T \in \text{range}(\Sigma_C)$.

The key innovation of this method is that the analyst is able to encode substantive knowledge in a restriction set $\mathcal{R}$ for the stacked vector $\beta = (\beta_0^\top, \beta_1^\top)^\top$ as a polyhedron or a finite union of polyhedra. [Boyarsky et al. \[2026\]](#) provide several such candidate restrictions: magnitude caps, effect ordering, sign restrictions, and budgets. The moments and the restriction define

$$\mathcal{F} = \{\beta : \Sigma_C \beta_a = d_a, a \in \{0, 1\}, \beta \in \mathcal{R}\}, \quad (2.5)$$

and the sensitivity interval is $[v_-, v_+]$ with

$$v_- = b_T + \inf_{\beta \in \mathcal{F}} \ell_T^\top(\beta_1 - \beta_0), \quad v_+ = b_T + \sup_{\beta \in \mathcal{F}} \ell_T^\top(\beta_1 - \beta_0). \quad (2.6)$$

In this paper, we additionally define $W = v_+ - v_-$ as the width of the identified set at the target. An infeasible program means the restriction contradicts the source moments. An unbounded program means the restriction leaves a direction that matters for the target uncontrolled. When W is a finite value it gives us a natural tool to characterize the generalizability of a particular set of experiments to a new target site.

Two more objects from [Boyarsky et al. \[2026\]](#) are used in this paper. The first is the reduced coordinates. Given a fixed orthonormal basis $B_C \in \mathbb{R}^{p \times r}$ of $\mathcal{V}_C$, write $\Omega = B_C^\top \Sigma_C B_C$ and $\xi_a = B_C^\top d_a$. The equality system in (2.5) is equivalent to $\Omega B_C^\top \beta_a = \xi_a$, which has full row rank when the smallest eigenvalue of Ω is positive. We assume this eigenvalue condition throughout, so that $\text{range}(\Sigma_C) = \mathcal{V}_C$ and $\ker(\Sigma_C) = \mathcal{V}_C^\perp$. The second component is the estimator as defined by Algorithm 1. This algorithm first estimates the five inputs $\Gamma = (b_T, \ell_T, \Sigma_C, d_0, d_1)$ with cross-fitted orthogonal scores, fits m_C through a joint site-probability classifier so that every fitted residual lies in $\mathcal{V}_C$ exactly, pools the scores, and solves each bound program once in the reduced coordinates. We call it the input-debiased estimator, in contrast to the one-step estimator also studied in [Boyarsky et al. \[2026\]](#). Every estimate reported in this paper uses it.

3 The site-selection problem

3.1 Data requirements for site-selection

We assume the analyst holds data from the S existing sites, which we call the source $\mathcal{S}$, and faces a candidate set $\mathcal{K} = \{1, \dots, K\}$ of prospective sites. For each candidate k we know the context profile $c^{(k)} \in \mathbb{R}^p$, which has same basis as the source profiles. This context is site-level information. For example, in the [Meager \[2019\]](#) dataset we use interest rates, program rules, administrative statistics. Often these data are available from public sources before any experiment is conducted. We also need some information about the candidate's unit-level covariate distribution. This can often come from a census, a survey, a prior, or the assumption that units there resemble units at a reference site. We do not have experimental results from the candidate sites, such as treatment assignments or outcome data, because no experiment has been conducted there.

We also state the target we care about as a set $\mathcal{T}$ of context profiles. This can be a single profile, a list of benchmark profiles, or a box around a forecast of the target environment. The design problem is

$$k^* \in \arg \min_{k \in \mathcal{K}} \sup_{c_T \in \mathcal{T}} W(c_T; \mathcal{S} \cup \{k\}), \quad (3.1)$$

the minimax width of the identified set over the targets, evaluated as if candidate k 's experiment had been run. When candidates carry costs, as is often the case in practice, the investigator can add a cost term to the objective or screen the pool by budget first. Everything below goes through

in either case because a cost enters separably in k . Choosing a site is a discrete choice that is only made once, and it should not depend on which single target profile turns out to be the relevant one. Hence, the minimax design is a natural encoding of this problem.

The main obstacle is that $W(c_T; \mathcal{S} \cup \{k\})$ depends on data that we do not have. The following two sections show how much we can determine via the candidate’s context alone.

3.2 What adding a site changes

Suppose the candidate experiment is run, with n_k observations and a randomized treatment within the site, and its data are pooled with the existing sites. Three inputs of the bound programs change.

First, the span. The centered profile span grows from $\mathcal{V}_C$ to

$$\mathcal{V}'_C = \mathcal{V}_C + \text{span}\{c^{(k)} - c_1\}, \quad (3.2)$$

so its dimension rises by one when $c^{(k)} - c_1 \notin \mathcal{V}_C$ and is unchanged otherwise. Notice that the span update is a function of the profiles alone. That is, we know it at “site choice” time.

Second, the moments. The pooled law now mixes in the candidate site, so Σ_C , d_a , m_C , and hence b_T and ℓ_T all move. Under Assumption 1 the new normal equations remain consistent with the same β_a . In Section 5 we show that their entire effect on the feasible set is captured by one scalar per arm.

Third, the equality constraint rank. Whether the extra span direction becomes an extra rank of Σ_C depends on covariate overlap. The matrix Σ_C averages the conditional covariance of C given X , and a site contributes context variation only at covariate values where units could have come from more than one site. A candidate whose covariate support is disjoint from the pool can contribute profile diversity but no conditional variation. In that case site membership is a deterministic function of X on the candidate’s support, the residual $\tilde{C}$ vanishes there, and the added direction is not identified. We therefore maintain the eigenvalue condition of Section 2 for the augmented pool: the covariance of $B'_C \tilde{C}$ has a positive smallest eigenvalue, where B'_C is a basis of $\mathcal{V}'_C$. In words, the candidate must overlap the pool in unit-level covariates, in the same way the target must.

Throughout, the target objects (Q_T, c_T) are unchanged by the addition. Only the source pool grows.

4 The geometry of adding a site

This section describes what identification looks like in the space of site profiles. Every identification statement in the selection problem reduces to a statement about the affine hull

$$\text{aff}(\mathcal{S}) = \left\{ \sum_{s=1}^S w_s c_s : \sum_{s=1}^S w_s = 1 \right\}, \quad (4.1)$$

the set of affine combinations of the observed profiles. Its direction space is exactly $\mathcal{V}_C$, the span of the centered profiles from (2.4).

The first step relates the target loading to the hull. The loading $\ell_T = c_T - \mathbb{E}_{Q_T}[m_C(X)]$ subtracts from the target profile a point that is always a mixture of observed profiles, because $m_C(x) = \sum_s \mathbb{P}(C = c_s \mid X = x) c_s$ is a convex combination of them at every x . The unidentified part of the loading is therefore the part of the target profile itself that lies outside the hull.

Lemma 1 (The loading and the affine hull). *Under Assumption 1 and the eigenvalue condition of Section 2, $\ell_T \in \text{range}(\Sigma_C)$ if and only if $c_T \in \text{aff}(\mathcal{S})$. Moreover, for the orthogonal projection $\Pi^\perp$ onto $\mathcal{V}_C^\perp$,*

$$\Pi^\perp \ell_T = \Pi^\perp (c_T - c_1), \quad \left\| \Pi^\perp \ell_T \right\|_2 = \text{dist}(c_T, \text{aff}(\mathcal{S})). \quad (4.2)$$

See Section A for all proofs. Lemma 1 turns the point-identification condition of the companion paper into a statement an investigator can check from the profiles alone. The pooled experiments determine the target effect exactly when the target’s context is an affine combination of the observed contexts. The relevant set is the affine hull, not the convex hull. A target outside the convex hull but inside the affine hull is still point identified, because the fixed effects are linear in context, so the model allows linear extrapolation along observed directions. Identification fails only in a direction that no combination of observed sites reaches.

Without restrictions, partial identification in the response model reduces to a dichotomy. [Bo-yarsky et al. \[2026\]](#) formalize this.

Proposition 1 (Unrestricted dichotomy). *Let $\mathcal{R} = \mathbb{R}^{2p}$. Under the conditions of Lemma 1,*

$$W(c_T; \mathcal{S}) = \begin{cases} 0 & \text{if } c_T \in \text{aff}(\mathcal{S}), \\ +\infty & \text{otherwise.} \end{cases} \quad (4.3)$$

A candidate site enters this geometry through one object only, the direction its profile adds to the span.

Proposition 2 (Span expansion is affine-hull expansion). *Let u_k be the component of $c^{(k)} - c_1$ orthogonal to $\mathcal{V}_C$. Then $\mathcal{V}'_C$ of (3.2) satisfies $\dim \mathcal{V}'_C = r + 1$ if and only if $u_k \neq 0$, which holds if and only if $c^{(k)} \notin \text{aff}(\mathcal{S})$. In that case $\mathcal{V}'_C^\perp = \mathcal{V}_C^\perp \cap \{u_k\}^\perp$, and under the augmented eigenvalue condition of Section 3.2 the augmented covariance satisfies $\text{range}(\Sigma'_C) = \mathcal{V}'_C$ and $\ker(\Sigma'_C) = \mathcal{V}'_C^\perp$. If instead $c^{(k)} \in \text{aff}(\mathcal{S})$, the span, the range, and the kernel are unchanged.*

Together with Proposition 1, this gives the unrestricted selection rule. A candidate collapses an infinite width at target c_T to zero exactly when its profile lies outside the hull in the direction of the target’s offset from the hull. Only the direction of u_k enters. How far outside the hull the candidate sits does not change what becomes identified, because a fixed-effects model pins the same coefficient combination from a small step along a direction as from a large one. Distance along the new direction affects the precision of the added constraint, through the context variance the new site contributes. A farther site therefore improves precision but not identification.

Restrictions replace the dichotomy with finite widths. In that case the distance from the target to the hull is the right continuous measure of what remains unidentified. The simplest statement uses a cap on the size of the treatment shift $\delta = \beta_1 - \beta_0$.

Proposition 3 (Width equals distance under a norm cap). *Let the restriction be $\|\delta\|_2 \leq \tau$ with no restriction on β_0 , let $\bar{\delta}$ be the minimum-norm solution of $\Sigma_C \delta = d_1 - d_0$, and assume $\tau \geq \|\bar{\delta}\|_2$. Under the conditions of Lemma 1,*

$$W(c_T; \mathcal{S}) = 2\sqrt{\tau^2 - \|\bar{\delta}\|_2^2} \text{dist}(c_T, \text{aff}(\mathcal{S})). \quad (4.4)$$

The same formula holds for the augmented pool with $\text{aff}(\mathcal{S} \cup \{k\})$ in place of $\text{aff}(\mathcal{S})$, so under this restriction the minimax problem (3.1) says to choose the candidate whose profile, added to the hull, leaves no target in $\mathcal{T}$ far from it. The relation between the moment geometry and the context geometry is exact, with no side condition beyond overlap, because Lemma 1 identifies the unrestricted direction of the loading with the target’s offset from the hull.

5 Scoring candidates under restrictions

Applied bounds are computed under substantive restrictions, so the selection rule must handle a polyhedral $\mathcal{R}$, where widths are finite and the dichotomy of Proposition 1 no longer holds. The key structural fact is that the candidate's entire effect on the feasible set passes through two scalars.

Proposition 4 (Adding a site adds two equality constraints). *Suppose Assumption 1 holds for the pool and for candidate k , and the eigenvalue condition holds for the augmented pool. Let $\mathcal{F}'$ denote the feasible set (2.5) built from the augmented moments. If $c^{(k)} \notin \text{aff}(\mathcal{S})$, then, with u_k normalized to unit length,*

$$\mathcal{F}' = \mathcal{F} \cap \{\beta : u_k^\top \beta_a = y_{a,k}, a \in \{0, 1\}\}, \quad y_{a,k} = u_k^\top \beta_a, \quad (5.1)$$

where $y_{a,k}$ is identified by the augmented pool but not by $\mathcal{S}$. If $c^{(k)} \in \text{aff}(\mathcal{S})$, then $\mathcal{F}' = \mathcal{F}$. The bounds (2.6) built from the augmented moments equal the bounds built from the base objective (b_T, ℓ_T) over $\mathcal{F}'$: the changes in b_T and ℓ_T induced by re-pooling shift both endpoints by the same amount tied to the re-decomposition of ψ_T , and the target effect interval is unchanged by them.

Proposition 4 is the basis of the design rule. The direction u_k is computable from the profiles before the experiment. The scalars $y_{0,k}$ and $y_{1,k}$ are what the candidate experiment reveals: the projection of each arm's fixed effects on the new direction. Everything else about the candidate's data (its sample size, its noise level, and its covariate mix) affects only how precisely those two scalars are estimated. It does not affect the identified set.

The predicted interval. Define the design-time width map

$$W_k(c_T; y_0, y_1) = \text{width of (2.6) over } \mathcal{F} \cap \{u_k^\top \beta_a = y_a, a \in \{0, 1\}\}, \quad (5.2)$$

a pair of linear programs the investigator can solve before the experiment, for any conjectured (y_0, y_1) . Before the experiment we evaluate it at a forecast. The neutral forecast extends the pool's own fit. The minimum-norm solution $\hat{\beta}_a$ of the estimated equality system sets the unidentified coordinates to zero, so its forecast is $\hat{y}_a = u_k^\top \hat{\beta}_a = 0$. Any substantive prior that maps the candidate's context to expected outcomes can replace it. We call $W_k(c_T; \hat{y}_0, \hat{y}_1)$ the predicted width and use it, maximized over $\mathcal{T}$, as the score in (3.1).

Pricing the forecast error. The prediction is wrong only to the extent that the forecast is wrong. The sensitivity follows from a standard envelope argument. At a bound with a unique optimizer and dual, the derivative of the bound in y_a is the multiplier on the added row, so for small errors

$$v_j(y) \approx v_j(\hat{y}) + \sum_{a \in \{0,1\}} \lambda_{j,a}^u (y_a - \hat{y}_a), \quad j \in \{-, +\}, \quad (5.3)$$

where $\lambda_{j,a}^u$ is bound j 's dual variable on the constraint $u_k^\top \beta_a = y_a$. This is the same envelope argument that the inference theory of the companion paper uses, applied to the added rows. An investigator who can bound the forecast error, say $|y_a - \hat{y}_a| \leq B$, obtains a range of predicted intervals by adding $\sum_a (|\lambda_{-,a}^u| + |\lambda_{+,a}^u|) B$ to the predicted width, or exactly by re-solving (5.2) on a grid of y values, since the map is piecewise linear in y . In our simulations and in the case study the ranking of candidates is stable across these variants, because the ranking depends on which direction is resolved and not on the forecast level.

Three kinds of candidate. The three categories from the companion paper’s leave-one-out exercise also describe what a candidate can do.

- *Identification.* If $c^{(k)} \notin \text{aff}(\mathcal{S})$, the candidate adds the equality constraints (5.1). This is the only way a candidate changes the identified set, and it is checkable from profiles alone.
- *Precision.* If $c^{(k)} \in \text{aff}(\mathcal{S})$, the identified set is unchanged, and the candidate’s value is statistical. More context variation inside $\mathcal{V}_C$ tightens the estimates and the standard errors of the same bounds. Interior sites add precision but not identification.
- *Feasibility.* A candidate can also restore or destroy feasibility. If the pool’s program is infeasible under $\mathcal{R}$, no addition restores it, since $\mathcal{F}' \subseteq \mathcal{F}$ under the model. But an addition whose data conflict with the model can make a feasible program infeasible. An infeasible augmented program then signals that conflict, not a property of the candidate. Section 6 takes this up.

Scoring procedure. The rule we use in Sections 8 and 9 has four steps.

1. For each candidate, test $c^{(k)}$ against $\text{aff}(\mathcal{S})$.
2. For each candidate that expands the hull, solve (5.2) at the forecast over $\mathcal{T}$.
3. Report the minimax score, the resolved direction, and the dual prices of the forecast.
4. Break ties, and rank interior candidates, by predicted precision at fixed identification.

The whole computation reuses the base-pool estimate $\hat{\Gamma}$ and adds two rows per candidate, so scoring a pool of K candidates costs K pairs of linear programs per target beyond the base fit.

6 When adding a site can hurt

One might hope that new evidence can only help, so that the design problem is only about how much a candidate helps. Under the maintained model this is true.

Theorem 1 (Monotone improvement under the model). *Suppose Assumption 1 holds for the pool and for candidate k , the augmented eigenvalue condition holds, and the true coefficient vector satisfies $\beta \in \mathcal{R}$. Then $\mathcal{F}' \subseteq \mathcal{F}$, both sets contain β , and for every target c_T ,*

$$W(c_T; \mathcal{S} \cup \{k\}) \leq W(c_T; \mathcal{S}), \quad (6.1)$$

with equality for every target when $c^{(k)} \in \text{aff}(\mathcal{S})$. In particular the augmented program is feasible and its interval still contains the true target effect ψ_T .

The assumptions of Theorem 1 show where it can fail. Every assumption is a modeling commitment about the new site. The new site must share the response surfaces g_a and the fixed effects β_a of (2.1). A site that does not share them, e.g., because its implementation or its measurement differs, moves the pooled normal equations. The normal equations then average conflicting site-level means, and the identified slice of coefficient space translates to a compromise that fits no site exactly. The restriction set then cuts the translated slice at a different cross-section, and the width can move either way.

A two-site example. Take one context coordinate pair, $p = 2$, and two source sites with profiles $c_1 = (0, 0)$ and $c_2 = (1, 1)$, the two-site configuration [Boyarsky et al. \[2026\]](#) use to illustrate partial identification. The pool identifies one combination of the treatment shift, $\delta_1 + \delta_2 = 2$ say, and leaves the direction $(1, -1)$ free. Impose the box $\delta \in [0, 2] \times [-1, 1]$ and take a target that loads on δ_1 alone. The feasible segment is $\{(\delta_1, 2 - \delta_1) : \delta_1 \in [1, 2]\}$ and the width is 1. Now add a third site on the same line through the profiles whose experiment implies $\delta_1 + \delta_2 = 0$, in conflict with the pool. Pooling the two inconsistent moment conditions with equal weight moves the identified value of $\delta_1 + \delta_2$ to the compromise 1. The new feasible segment is $\{(\delta_1, 1 - \delta_1) : \delta_1 \in [0, 2]\}$ and the width is 2. A rank-preserving addition doubled the width without changing the restriction set. Panel (A) of Figure 2 draws this example.

What conflict corrupts first. Width is not the right measure of the damage. The translated slice can just as easily cut a narrower cross-section of the restriction set, in which case the conflicted pool reports a tighter interval around a wrong location. The precise statement concerns whether the interval covers the true effect. The translation moves the point the pool identifies, so the targets that suffer first are the point-identified ones. Their intervals have width zero, so any movement removes the true effect from the interval. A wide interval can still cover the true effect after a small conflict. A point prediction cannot. In the simulation of Section 8 the conflicted addition changes the worst-case width by only a few percent in either direction, while the population interval stops covering the true effect at the span-identified targets in every replicate. An analyst who sees a surprisingly tight interval after pooling a new site should treat it as a sign of conflict rather than as an improvement.

Screens. Two checks separate refining additions from corrupting ones.

- *Rank screen, before the experiment.* Test $c^{(k)} \notin \text{aff}(\mathcal{S})$. A candidate that passes adds the equality constraints (5.1) and, under the model, cannot widen anything (Theorem 1). A candidate that fails changes no identified quantity, so any movement of the estimated set after adding it, beyond sampling noise, is itself evidence of conflict.
- *Compatibility screen, after the experiment.* The candidate’s data overidentify the in-span coordinates: the projections $B_C^\top \beta_a$ are estimated by the pool and, once the experiment runs, by the candidate site jointly with the pool. Compare the pool’s minimum-norm fit with the augmented fit on the in-span coordinates, in units of their standard errors. A large discrepancy indicates the conflict mechanism above. The check is an overidentification test of the kind used to detect an invalid instrument. The forecast version of this screen, before the experiment, compares the candidate’s predicted outcomes under the pool’s fit with whatever prior information exists about outcomes there.

When a screen fails, the analyst should report both analyses, the bounds with and without the new site, together with the discrepancy between them. Dropping the new site assumes that the pool is right. Pooling assumes that the model is right. The screens make the disagreement visible so the analyst can decide which assumption to keep.

7 Site selection for omitted moderators

The fixed-effects response model is a statement about which site-level variables modify the response. It can fail because a relevant moderator is omitted from C . Omitted-variable bias is especially

useful here because it separates two quantities that are often conflated: the strength of the omitted moderator in the treatment effect and the extent to which the moderator is imbalanced between the source sites and the target.

Let $U \in \mathbb{R}^q$ be a site-level moderator with profile u_s at site s . For this section, suppose the treatment-effect contrast obeys the augmented response model

$$\mathbb{E}_P[Y(1) - Y(0) \mid X, C, U] = g_\Delta(X) + C^\top \theta + U^\top \gamma, \quad \theta = \beta_1 - \beta_0, \quad (7.1)$$

where $\gamma \in \mathbb{R}^q$ is the effect of the omitted moderator. The analysis in the rest of the paper fits the model without U . Define

$$m_U(x) = \mathbb{E}_P[U \mid X = x], \quad \Sigma_{CU} = \mathbb{E}_P[\text{Cov}(C, U \mid X)], \quad \ell_{U,T} = u_T - \mathbb{E}_{X \sim Q_T}[m_U(X)]. \quad (7.2)$$

The target now has an omitted-moderator profile u_T . These quantities need not be observed; they index the sensitivity analysis.

Proposition 5 (Omitted-moderator bias). *Let θ^C solve the normal equations from fitting the response model without U , so that*

$$\Sigma_C \theta^C = d_1 - d_0. \quad (7.3)$$

Under (7.1),

$$d_1 - d_0 = \Sigma_C \theta + \Sigma_{CU} \gamma. \quad (7.4)$$

If $c_T \in \text{aff}(\mathcal{S})$, then the target value reported by the misspecified model, $\psi_T^C = b_T + \ell_T^\top \theta^C$, differs from the true target effect by

$$\psi_T - \psi_T^C = (\ell_{U,T} - \ell_T^\top \Sigma_C^\dagger \Sigma_{CU})^\top \gamma. \quad (7.5)$$

The two terms in (7.5) have a direct interpretation. The loading $\ell_{U,T}$ is the source–target difference in the omitted moderator, while $\ell_T^\top \Sigma_C^\dagger \Sigma_{CU}$ is the component that the fitted context coefficients absorb because U is associated with C in the source pool. The formula is invariant to the choice of solution θ^C because $\ell_T \in \text{range}(\Sigma_C)$ when the target is in the affine hull.

The same decomposition gives a new role to sites that are interior in the observed context geometry. Let a candidate satisfy $c^{(k)} = \sum_s w_s c_s$ with $\sum_s w_s = 1$, and define its omitted-profile contrast

$$d_{U,k} = u_k - \sum_s w_s u_s. \quad (7.6)$$

The weights may be negative, as they are for affine, rather than convex, combinations. If $d_{U,k} \neq 0$, the candidate is outside the affine hull of the existing profiles in the augmented (C, U) space even though it is inside the affine hull in C alone. It therefore tests a direction that the maintained model treats as irrelevant.

Theorem 2 (Refinement under an omitted moderator). *Suppose the augmented response model (7.1) holds for the source pool and candidate k . Let ρ_k denote the population contrast between the candidate’s conditional treatment effect and the same affine combination of source-site conditional treatment effects. Then, for every x ,*

$$\begin{aligned} \rho_k &= \mathbb{E}[Y(1) - Y(0) \mid X = x, C = c^{(k)}, U = u_k] \\ &\quad - \sum_s w_s \mathbb{E}[Y(1) - Y(0) \mid X = x, C = c_s, U = u_s] = d_{U,k}^\top \gamma. \end{aligned} \quad (7.7)$$

Consider an omitted-moderator sensitivity set

$$\mathcal{F}_U = \{(\theta, \gamma) : \Sigma_C \theta + \Sigma_{CU} \gamma = d_1 - d_0, (\theta, \gamma) \in \mathcal{R}_U\}, \quad (7.8)$$

and the target functional

$$\psi_T(\theta, \gamma) = b_T + \ell_T^\top \theta + \ell_{U,T}^\top \gamma. \quad (7.9)$$

After the candidate experiment is observed, the feasible set is

$$\mathcal{F}_U^{(k)} = \mathcal{F}_U \cap \{(\theta, \gamma) : d_{U,k}^\top \gamma = \rho_k\}. \quad (7.10)$$

If the true (θ, γ) belongs to $\mathcal{F}_U$, then the corresponding identified-set width satisfies, for every target,

$$W_U(c_T, u_T; \mathcal{S} \cup \{k\}) \leq W_U(c_T, u_T; \mathcal{S}). \quad (7.11)$$

If $d_{U,k} = 0$, then $\rho_k = 0$ and the candidate adds no omitted-moderator restriction, so the two widths are equal. If $d_{U,k} \neq 0$, the candidate can strictly reduce the width whenever the restriction $d_{U,k}^\top \gamma = \rho_k$ cuts a target-relevant direction in $\mathcal{F}_U$.

The theorem is the omitted-variable analogue of Theorem 1. Under the maintained model, a candidate outside the observed affine hull is valuable because it identifies a new context direction. Under the augmented model, an interior candidate is valuable when it is outside the affine hull after the omitted moderator is included. The design criterion is therefore to seek sites for which $d_{U,k}$ is large in directions of γ that matter for $\ell_{U,T}$. If U is not measured, this direction must be supplied by a proxy, substantive prior, or sensitivity range. Without such information, an unrestricted omitted moderator remains unconstrained by any site-selection rule based on C alone.

8 Simulations

We check the theory with the estimator a user would run, not only with population calculations. Both experiments generate multi-site data from the fixed-effects model (2.1), estimate the five bound inputs with the rank consistent estimator of [Boyarsky et al. \[2026\]](#), and solve the estimated bound programs. The replication code reproduces every number and figure.

8.1 Design

The context space has $p = 6$ coordinates. Six existing sites carry fixed profiles whose centered span $\mathcal{V}_C$ has rank three, so three directions of the fixed effects are unidentified, the same mechanism as in the microcredit data. Each site contributes 500 observations with three unit-level covariates, a binary covariate, a discrete covariate whose distribution varies by site, and a Gaussian covariate. Treatment probabilities are known and vary by site between 0.35 and 0.65. An independent target covariate sample has 1,500 observations. The target region $\mathcal{T}$ holds fifteen fixed profiles, five inside the affine hull of the pool and ten with components outside it. The restriction is the per-arm box $|\beta_{a,j}| \leq 2.5$, which keeps every width finite and contains the truth with room to spare. Estimation uses four folds and correctly specified linear and multinomial logistic nuisance fits, so the experiments isolate the selection problem rather than nuisance error. Exact population moments and widths come from finite enumeration of the discrete covariates. Each experiment runs 40 replicates that redraw the data with the geometry fixed.

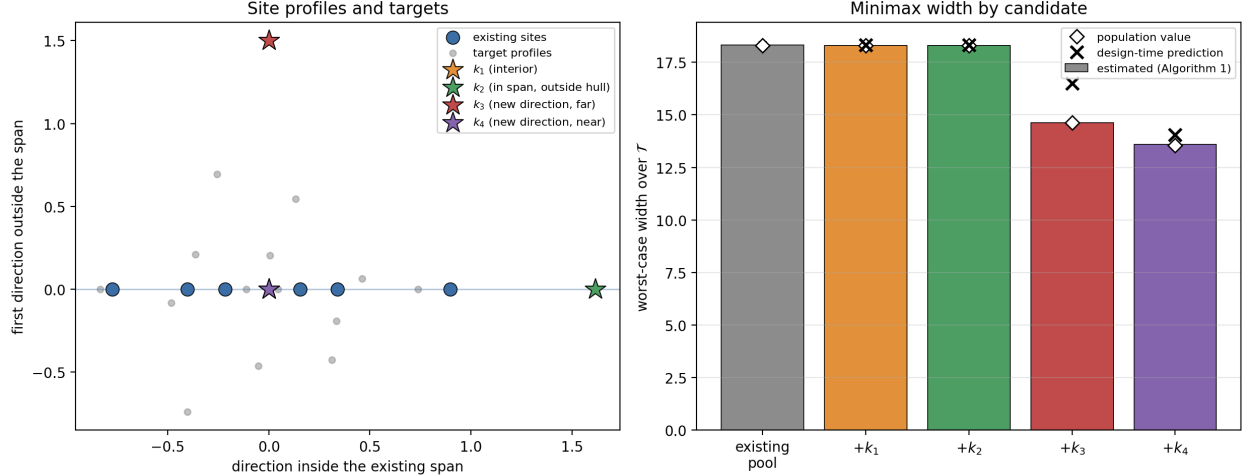


Figure 1. Site selection with estimated bounds. Left: the fixed geometry, projected onto one direction inside the existing span and the first direction outside it. Right: worst-case width over the target region for the existing pool and after each candidate, showing the mean and standard deviation of the estimated widths across 40 replicates, the exact population width, and the design-time prediction computed before the candidate experiment.

8.2 Selecting among four candidates

Four candidate sites cover the three categories of Section 5: k_1 inside the convex hull of the existing profiles, k_2 inside the affine hull but outside the convex hull, and k_3 and k_4 outside the affine hull along two different directions, k_3 far outside along the first and k_4 a short step along the second. For each candidate we compute the design-time prediction from the existing pool alone, then simulate the candidate experiment from the same model, refit the estimator on the seven-site pool, and record the realized worst-case width over $\mathcal{T}$.

Figure 1 reports the results. The population worst-case width is 18.31 for the existing pool and stays exactly 18.31 when k_1 or k_2 is added: sites inside the affine hull change nothing about identification, whether or not they are inside the convex hull. The two hull-expanding candidates cut the population worst-case width to 14.63 (k_3) and 13.54 (k_4), reductions of 20 and 26 percent. The estimated widths track the population values to within sampling noise in every cell, with mean estimates of 18.31, 18.31, 18.29, 14.63, and 13.59. The estimated selection picks k_4 , the population-optimal candidate, in all 40 replicates, and so does the design-time selection computed before the candidate data exist. The design-time predictions are conservative, 16.49 against a realized 14.63 for k_3 and 14.04 against 13.59 for k_4 , because the neutral forecast slices the restriction box through its widest cross-section, and they preserve the ranking.

The comparison between k_3 and k_4 makes the geometry of Section 4 concrete. The nearby candidate beats the distant one because the worst-case targets load more on the direction k_4 resolves than on the direction k_3 resolves. Distance to the hull does not enter the identified set. The identified set depends on which direction a candidate resolves and how much the target region loads on it.

8.3 The conflict mechanism

The second experiment tests Theorem 1 and its failure mode. Two profiles are reused, the in-span candidate k_2 and the hull-expanding candidate k_3 , and each is added twice: once with outcomes drawn from the maintained model, labels a_1 and a_3 , and once with the candidate site’s treated-arm

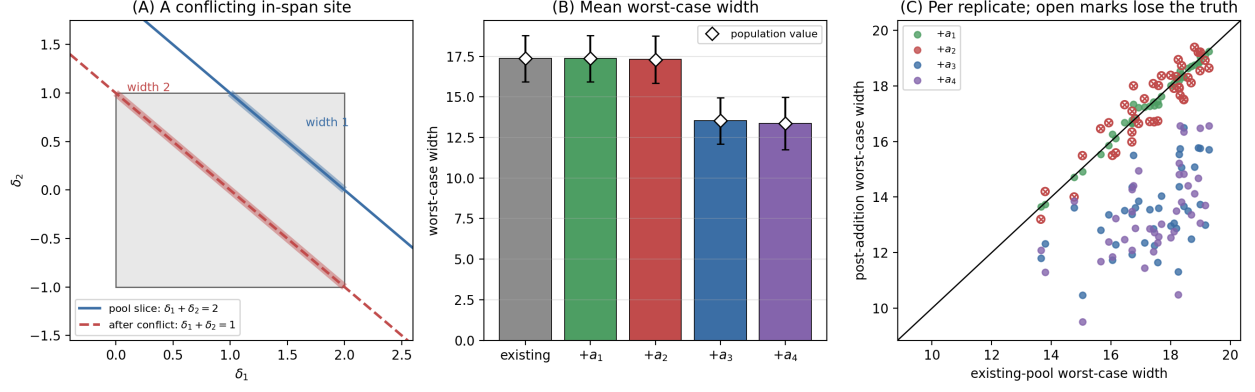


Figure 2. What a conflicting site does. (A) The two-site example of Section 6: a third site on the pool’s own line whose outcomes conflict with the pool moves the identified slice and doubles the width at the target. (B) Mean estimated worst-case width across 40 replicates with the exact population value marked. (C) Post-addition against existing-pool worst-case width by replicate. Open markers are replicates whose population identified set no longer covers the true effect at every target: all 40 replicates of the conflicted in-span addition a_2 , and none of the model-consistent additions.

mean shifted by a conflict of size 0.75 whose sign is drawn at random each replicate, labels a_2 and a_4 . The restriction box is re-centered at random each replicate so that it is generically asymmetric around the identified slice, which is when the translation mechanism of Section 6 can move widths. Population widths and truth coverage are computed exactly for every drawn box.

Figure 2 and its underlying table say three things. First, the model-consistent additions behave exactly as Theorem 1 requires: a_1 leaves the population worst-case width unchanged in all 40 replicates, a_3 strictly shrinks it in all 40, with reductions between 1.1 and 6.7, and neither ever loses the truth. Second, the conflicted in-span addition a_2 moves the population worst-case width in both directions, widening it in 45 percent of replicates and narrowing it in 55 percent, by up to 0.71 either way. The direction is unpredictable in advance, which is the point of the counterexample. Third, and most important, a_2 removes the true target effect from the population identified set at the span-identified targets in every replicate, while its worst-case width moves by at most a few percent. The width does not show the damage, and the interval no longer covers the true effect. The conflicted hull-expanding addition a_4 still shrinks widths here, and its bias lies in the directions that remain unidentified. This is why the compatibility screen of Section 6, and not the width, is the right check for it.

9 Case study: which microcredit experiment was worth running

The microcredit data let us test the design rule against experiments that have already been run. All seven country experiments were run, so we can hide one, score it as a candidate from its lending terms alone, and then reveal its data to check the prediction. This is the forward version of the leave-one-out exercise in Boyarsky et al. [2026], and it gives a design rule for future experiments a test it rarely gets, because the future data already exist.

9.1 Specification

The specification is the one fixed in the companion paper’s case study, and the replication code reproduces every number. The outcome is household revenues. The context basis holds nine coded terms: promotion, the annual interest rate, and the average loan size as a share of income, each

coded on the source pool’s rank scale, plus their squares and pairwise products, with every column scaled by its source-pool standard deviation. Unit-level covariates are the six standardized household variables. Estimation uses the rank consistent estimator with three folds, boosted outcome and density-ratio fits, and the regularized multinomial site classifier. The restriction ladder is the companion paper’s: a magnitude cap of $M = 888$ revenue units, one standard deviation of the outcome, on the control surface and on the treatment shift, then effect ordering at $\kappa = 1$, then the sign restrictions on the interest-rate terms of the treatment shift. We report the final rung. Coding is frozen at each target’s full six-site source pool, so that removing a source site changes only whose data enter the moments, and the held-out benchmarks are the covariate-adjusted design-based estimates with clustering at each country’s randomization unit.

For each target country, Bosnia and Morocco as in the companion paper, the exercise runs over the six five-site source pools. The missing sixth country is the candidate. Its coded profile is known and its data are hidden. At design time we run the rank test of Section 5, forecast the revealed scalars with the feasibility-aware neutral forecast, and solve the width map (5.2) for the predicted interval. The neutral forecast sets the new coordinate of the fixed effects to the value closest to zero that the pool’s own restrictions allow, which applies to two of the Bosnia pools, where the sign restrictions pin the coordinate away from zero. Then we reveal the candidate’s experiment, refit the estimator on all six source sites, and compare.

9.2 Bosnia: a predictable collapse

Every five-site source pool for Bosnia gives the same result (Table 1 and Figure 3). The centered profiles of five source sites span four directions against nine basis terms, the interval at the final rung is wide and unsigned, with widths between 3,525 and 7,291 revenue units, and no five-site pool supports a conclusion about Bosnia. The rank test, computed from lending terms alone, reports that every missing country expands the span. With five source sites and nine terms, each of the six countries’ coded profile lies outside the five-site hull. The predicted intervals collapse accordingly, with predicted reductions of 93 to 100 percent, and the realized six-site interval is $[37.1, 127.3]$, width 90, reductions of 97 to 99 percent. An analyst holding any five of these experiments, the lending terms of the sixth, and no outcome data from it would have been told that the sixth experiment removes almost all of the width that the restrictions leave, and the realized experiments confirm this.

The forecast cannot determine the location of the collapsed interval, and the rows differ in location. For four of the six pools the neutral forecast is an interior point of what the pool allows, the predicted intervals have widths of 121 to 286, and they sit just below and above zero. The revealed scalars then place the realized interval strictly above zero, which is the companion paper’s signed conclusion for Bosnia. For the other two pools, the ones missing India and Morocco, the pool’s sign restrictions pin the forecast coordinate to a boundary. The predicted intervals degenerate to near points, $[129.1, 138.6]$ and $\{67.5\}$, which overstates the collapse but is correct in one respect. Once that direction is resolved, the restrictions themselves nearly determine the answer, and they already place it above zero. The realized interval agrees in location and adds back the width, 90, that the boundary forecast removed.

9.3 Morocco: a correct ranking of modest gains

Morocco is the target the companion paper cannot sign, and the design rule explains why in advance. Even the full six-site source pool leaves most of Morocco’s loading outside the span of the source profiles, so no single candidate can deliver a collapse. What the rule can do is rank the six

target	candidate	base width	predicted	pred. gain	realized gain
Bosnia	Ethiopia	4,606	229	95.0%	98.0%
Bosnia	India	7,291	10	99.9%	98.8%
Bosnia	Mexico	4,109	235	94.3%	97.8%
Bosnia	Mongolia	4,171	286	93.2%	97.8%
Bosnia	Morocco	4,398	0	100.0%	97.9%
Bosnia	Philippines	3,525	121	96.6%	97.4%
Morocco	Bosnia	7,770	6,485	16.5%	17.4%
Morocco	Ethiopia	6,767	6,417	5.2%	5.2%
Morocco	India	8,074	6,482	19.7%	20.5%
Morocco	Mexico	8,425	6,480	23.1%	23.8%
Morocco	Mongolia	10,453	6,487	37.9%	38.6%
Morocco	Philippines	8,165	6,487	20.5%	21.4%

Table 1. Predicted and realized value of the missing experiment. Each row hides one country’s experiment from the target’s source pool. Base width is the width of the identified set at the final restriction rung with five source sites. Predicted is the width after adding the candidate, computed before its data from the coded lending terms and the feasibility-aware neutral forecast. Gains are relative to the base width; the realized gain uses the six-site interval. Widths are in revenue units.

candidates, and here the check is informative, because each base pool differs and each realized gain is a different number. The predicted gains run from 5.2 percent for Ethiopia to 37.9 percent for Mongolia. The realized gains run from 5.2 to 38.6 percent, and they match the predictions candidate by candidate to within about one percentage point (Table 1). The ranking is exact. Mongolia’s experiment, which has the highest interest rate in the pool and the lending environment most unlike the others, is the one that helps Morocco most. Ethiopia’s experiment, whose coded terms the other source sites nearly reproduce, adds almost nothing. An investigator with the five-country pool and the six candidates’ lending terms would have chosen Mongolia, and she would have been right, for the predicted reason. The rule also reports what none of the candidates fixes. Every predicted and realized interval for Morocco stays wide and unsigned. Strengthening the Morocco conclusion requires a site outside the observed hull in Morocco’s direction, not a sixth source site from this pool.

9.4 What the exercise does and does not show

Three caveats limit the claim. First, the exercise conditions on the companion paper’s specification: the coded basis, the frozen coding, and the restriction ladder. The predictions are exact statements about that specification, not about microcredit at large. Second, the predictions use estimated base-pool inputs, so they inherit sampling error; the near-exact match at Morocco partly reflects that base and augmented fits share five countries of data. Third, the exercise also tests Theorem 1. The realized additions never widened an interval, which is consistent with the maintained model, and the compatibility screen of Section 6 raises no conflict at the final rung for either target, since the augmented programs remain feasible. When a screen does fail, as it does on these data for tighter caps in the supplementary appendix of the companion paper, the right object to report is the disagreement itself.

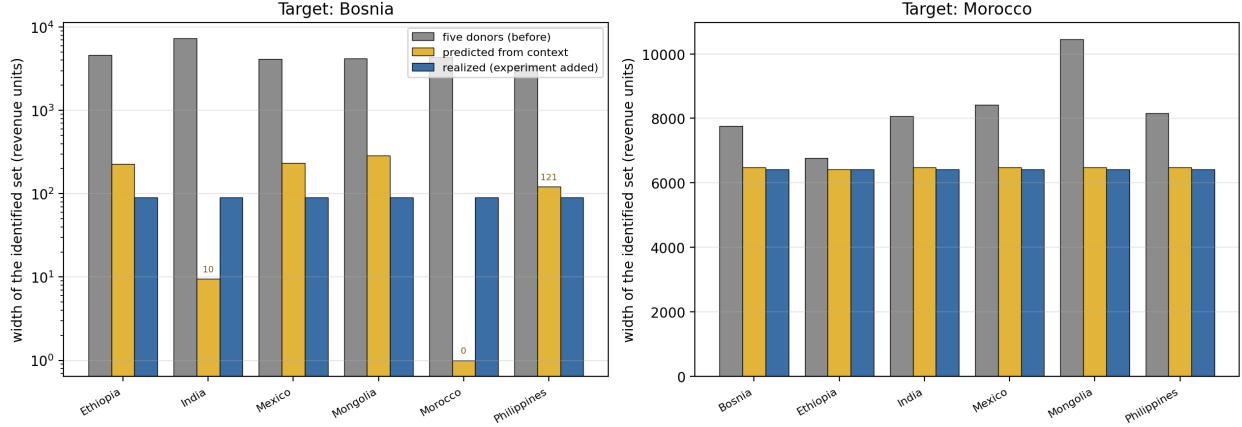


Figure 3. Predict, then check, on the microcredit experiments. For each target and each hidden country: the width of the identified set at the final restriction rung with five source sites, the width predicted at design time from the hidden country’s lending terms, and the realized width once its experiment is added. The Bosnia panel uses a logarithmic scale because its widths span two orders of magnitude, with the near-point predictions labeled.

10 Discussion

The sensitivity framework of [Boyarsky et al. \[2026\]](#) separates what pooled experiments identify from what must be assumed. This paper adds the design question. When the answer is a wide interval, where should the next experiment go? The question is answerable before the experiment to a larger degree than we expected. Identification gains are a property of the candidate’s context profile relative to the affine hull of the observed profiles. The bounds after the addition are computable in advance up to one revealed scalar per treatment arm, with the forecast error priced by the dual of the added constraint. Under the maintained model a new site can only refine the set, and when a new site instead moves conclusions the pool had pinned down, that movement is evidence against the pooled model, concentrated exactly where the pool claimed precision. The microcredit exercise suggests that these statements also hold in finite samples. Computed from lending terms with the companion paper’s estimator, the predictions matched the realized experiments.

Several extensions follow the same lines. Costs enter separably, so a budgeted rule ranks candidates by predicted gain per unit cost within the screened pool. Choosing several sites at once is a coverage problem. The identification gain of a set of candidates is the rank added by their profiles, which is a matroid rank function, so greedy selection has the usual submodular guarantee for the identification part. The width objective itself should be re-solved at the chosen set. When the candidate’s covariate distribution is unknown, the overlap condition of [Section 3.2](#) is the input most likely to fail, and a worst-case treatment of the candidate’s covariate law would make the precision part of the score conservative without changing the span part. Inference for the selected design, as opposed to inference for the bounds at a fixed design, remains open. The selection is a minimum of a maximum of estimated program values, and the tools in the companion paper cover each fixed program but not the selection event.

Two practical lessons follow. First, the design question depends on the choice of context basis. Every statement in this paper is relative to the chosen context basis, and a candidate that expands the hull in one coding can be interior in another. The companion paper’s sweeps show how much coding moves identification geometry, and a selection exercise should be run over the same codings the final analysis will report. Second, the most useful output of the rule can be a negative result. At Morocco the rule reports that no available candidate delivers a signed conclusion, and that the

useful experiment is one nobody has run, in a lending environment beyond the observed hull in Morocco’s direction. A design rule that can report that no available candidate helps, and which direction would help, gives a concrete request for the next trial.

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A Proofs

Throughout the appendix, Assumption 1 holds for every site that appears, including candidate sites unless the text says otherwise, and the eigenvalue condition of Section 2 holds for the pool and, where an augmented pool appears, for the augmented pool. We write Π_V and $\Pi^\perp$ for the orthogonal projections onto $\mathcal{V}_C$ and $\mathcal{V}_C^\perp$, and $\beta^\circ = (\beta_0^\circ, \beta_1^\circ)$ for the true coefficient vector of (2.1). Recall from (2.4) that $\mathcal{V}_C$ is the span of the centered site profiles and that $\text{aff}(\mathcal{S}) = c_1 + \mathcal{V}_C$.

Two facts from Boyarsky et al. [2026] are used repeatedly. First, the normal equations: under Assumption 1, for any pooled source law whose sites obey (2.1) with the same coefficients,

$$\text{Cov}(C, R_a \mid X) = \text{Cov}(C \mid X) \beta_a^\circ \quad \text{almost surely, so} \quad d_a = \Sigma_C \beta_a^\circ. \quad (\text{A.1})$$

Second, under the eigenvalue condition, $\text{range}(\Sigma_C) = \mathcal{V}_C$ and $\ker(\Sigma_C) = \mathcal{V}_C^\perp$, so the solution set of the equality system in (2.5) is $\{\beta_a : \Sigma_C \beta_a = d_a\} = \beta_a^\circ + \mathcal{V}_C^\perp$.

A.1 Proof of Lemma 1

Proof Write $p_s(x) = \mathbb{P}_P(C = c_s \mid X = x)$, so $m_C(x) = \sum_{s=1}^S p_s(x) c_s$ with $p_s(x) \geq 0$ and $\sum_s p_s(x) = 1$ for P -almost every x , hence for Q_T -almost every x by overlap. Then $q := \mathbb{E}_{Q_T}[m_C(X)] = \sum_s \bar{w}_s c_s$ with $\bar{w}_s = \mathbb{E}_{Q_T}[p_s(X)] \geq 0$ and $\sum_s \bar{w}_s = 1$, so $q \in \text{conv}\{c_1, \dots, c_S\} \subseteq \text{aff}(\mathcal{S})$. Since $\text{aff}(\mathcal{S}) = c_1 + \mathcal{V}_C$, we have $q - c_1 \in \mathcal{V}_C$ and therefore

$$\Pi^\perp \ell_T = \Pi^\perp (c_T - q) = \Pi^\perp (c_T - c_1) - \Pi^\perp (q - c_1) = \Pi^\perp (c_T - c_1).$$

The distance from a point to the affine set $c_1 + \mathcal{V}_C$ is the norm of the component of the offset orthogonal to $\mathcal{V}_C$, which gives (4.2). Finally, under the eigenvalue condition $\text{range}(\Sigma_C) = \mathcal{V}_C$, so $\ell_T \in \text{range}(\Sigma_C)$ if and only if $\Pi^\perp \ell_T = 0$, if and only if $\Pi^\perp (c_T - c_1) = 0$, if and only if $c_T \in c_1 + \mathcal{V}_C = \text{aff}(\mathcal{S})$. $\square$

A.2 Proof of Proposition 1

Proof With $\mathcal{R} = \mathbb{R}^{2p}$ the feasible set is the product of the two equality solution sets, $\mathcal{F} = \prod_{a \in \{0,1\}} (\beta_a^\circ + \mathcal{V}_C^\perp)$, which contains β° . For $\beta_a = \beta_a^\circ + z_a$ with $z_a \in \mathcal{V}_C^\perp$,

$$\ell_T^\top (\beta_1 - \beta_0) = \ell_T^\top (\beta_1^\circ - \beta_0^\circ) + (\Pi^\perp \ell_T)^\top (z_1 - z_0),$$

because $z_1 - z_0 \in \mathcal{V}_C^\perp$. If $c_T \in \text{aff}(\mathcal{S})$, then $\Pi^\perp \ell_T = 0$ by Lemma 1, the objective is constant on $\mathcal{F}$, and both endpoints equal ψ_T , so the width is zero. If $c_T \notin \text{aff}(\mathcal{S})$, then $\Pi^\perp \ell_T \neq 0$. Taking $z_1 = t \Pi^\perp \ell_T$ and $z_0 = 0$ with $t \rightarrow \pm\infty$ drives the objective to $\pm\infty$, so the width is infinite. $\square$

A.3 Proof of Proposition 2

Proof Write $u = \Pi^\perp (c^{(k)} - c_1)$, so $c^{(k)} - c_1 = \Pi_V (c^{(k)} - c_1) + u$ and

$$\mathcal{V}'_C = \mathcal{V}_C + \text{span}\{c^{(k)} - c_1\} = \mathcal{V}_C + \text{span}\{u\}.$$

If $u = 0$ the two spans coincide. If $u \neq 0$ the sum is orthogonal and $\dim \mathcal{V}'_C = r + 1$; moreover $u = 0$ exactly when $c^{(k)} \in c_1 + \mathcal{V}_C = \text{aff}(\mathcal{S})$, which proves the equivalences. For the complement, $x \in \mathcal{V}'_C^\perp$ if and only if $x \perp \mathcal{V}_C$ and $x \perp u$, that is $\mathcal{V}'_C^\perp = \mathcal{V}_C^\perp \cap \{u\}^\perp$. Finally, every residual $C - m'_C(X)$ of the augmented pool lies in $\mathcal{V}'_C$, because $m'_C(x)$ is a convex combination of the augmented profiles, so $\text{range}(\Sigma'_C) \subseteq \mathcal{V}'_C$ always, and the augmented eigenvalue condition states that $B_C'^\top \Sigma'_C B'_C$ is positive definite, which forces $\text{range}(\Sigma'_C) = \mathcal{V}'_C$ and, by symmetry of Σ'_C , $\ker(\Sigma'_C) = \mathcal{V}'_C^\perp$. $\square$

A.4 Proof of Proposition 3

Proof Only the shift $\delta = \beta_1 - \beta_0$ enters the objective and the restriction, and the pair of equality systems constrains it through $\Sigma_C \delta = d_1 - d_0$: given any such δ , choosing β_0 with $\Sigma_C \beta_0 = d_0$ and $\beta_1 = \beta_0 + \delta$ satisfies both systems. The solution set for the shift is $\bar{\delta} + \mathcal{V}_C^\perp$, where $\bar{\delta}$ is the minimum-norm solution and lies in $\mathcal{V}_C$. For $\delta = \bar{\delta} + \nu$ with $\nu \in \mathcal{V}_C^\perp$, orthogonality gives $\|\delta\|_2^2 = \|\bar{\delta}\|_2^2 + \|\nu\|_2^2$, so the cap $\|\delta\|_2 \leq \tau$ is exactly $\|\nu\|_2 \leq \rho_\tau := (\tau^2 - \|\bar{\delta}\|_2^2)^{1/2}$, and the objective is $\ell_T^\top \delta = \ell_T^\top \bar{\delta} + (\Pi^\perp \ell_T)^\top \nu$.

By the Cauchy–Schwarz inequality, the supremum and infimum of $(\Pi^\perp \ell_T)^\top \nu$ over the ball of radius ρ_τ in $\mathcal{V}_C^\perp$ are $\pm \rho_\tau \|\Pi^\perp \ell_T\|_2$, attained at $\nu = \pm \rho_\tau \Pi^\perp \ell_T / \|\Pi^\perp \ell_T\|_2$ when $\Pi^\perp \ell_T \neq 0$ and trivially when it is zero. The width is therefore $2\rho_\tau \|\Pi^\perp \ell_T\|_2$, and Lemma 1 identifies the last factor with $\text{dist}(c_T, \text{aff}(\mathcal{S}))$. $\square$

A.5 Proof of Proposition 4

Proof Apply (A.1) to the augmented pool: because the candidate site obeys (2.1) with the same coefficients, $d'_a = \Sigma'_C \beta_a^\circ$, so the augmented equality solution set is $\beta_a^\circ + \ker(\Sigma'_C)$, and by Proposition 2 this kernel is $\mathcal{V}_C^\perp \cap \{u_k\}^\perp$ in the hull-expanding case and $\mathcal{V}_C^\perp$ in the rank-preserving case.

In the rank-preserving case the solution sets coincide with the pool's, so $\mathcal{F}' = \mathcal{F}$ after intersecting with $\mathcal{R}$. In the hull-expanding case, for $\beta_a = \beta_a^\circ + z_a$ with $z_a \in \mathcal{V}_C^\perp$ we have $u_k^\top \beta_a = u_k^\top \beta_a^\circ + u_k^\top z_a$, and $z_a \in \{u_k\}^\perp$ exactly when $u_k^\top \beta_a = u_k^\top \beta_a^\circ = y_{a,k}$. Hence

$$\beta_a^\circ + \ker(\Sigma'_C) = (\beta_a^\circ + \mathcal{V}_C^\perp) \cap \{\beta_a : u_k^\top \beta_a = y_{a,k}\},$$

and intersecting both arms' sets with $\mathcal{R}$ gives (5.1). The scalar $y_{a,k}$ is a linear functional of $B_C'^\top \beta_a^\circ$, which the augmented pool identifies through its normal equations because $u_k \in \mathcal{V}_C'$; it is not identified by the pool alone because moving β_a along $u_k \in \mathcal{V}_C^\perp$ leaves every pool moment unchanged.

For the last claim, compare the augmented-pool endpoint programs, which use (b'_T, ℓ'_T) , with the base-objective programs over $\mathcal{F}'$. Write $q = \mathbb{E}_{Q_T}[m_C(X)]$ and $q' = \mathbb{E}_{Q_T}[m'_C(X)]$ for the two mixture points. As in the proof of Lemma 1, q lies in $\text{aff}(\mathcal{S})$ and q' in $\text{aff}(\mathcal{S} \cup \{k\})$, so $\ell_T - \ell'_T = q' - q \in \mathcal{V}_C'$, and for every $\beta \in \mathcal{F}'$, writing $\delta = \beta_1 - \beta_0 = \delta^\circ + \nu$ with $\nu \in \mathcal{V}_C'^\perp$,

$$(\ell_T - \ell'_T)^\top \delta = (\ell_T - \ell'_T)^\top \delta^\circ$$

is constant on $\mathcal{F}'$. Both decompositions represent the same target effect, $\psi_T = b_T + \ell_T^\top \delta^\circ = b'_T + \ell'^\top_T \delta^\circ$, so $b'_T - b_T = (\ell_T - \ell'_T)^\top \delta^\circ$, and for every $\beta \in \mathcal{F}'$,

$$b'_T + \ell'^\top_T \delta = b_T + (\ell_T - \ell'_T)^\top \delta^\circ + \ell_T^\top \delta - (\ell_T - \ell'_T)^\top \delta = b_T + \ell_T^\top \delta.$$

The two programs have the same objective value at every feasible point, hence the same endpoints. $\square$

A.6 Proof of Proposition 5

Proof Write $\tilde{C} = C - m_C(X)$ and $\tilde{U} = U - m_U(X)$. Randomization and (7.1) imply

$$\text{Cov}(C, R_1 - R_0 \mid X) = \text{Cov}(C \mid X)\theta + \text{Cov}(C, U \mid X)\gamma.$$

Taking expectations gives (7.4). Let $q = \Sigma_{CU}\gamma$. Every solution of (7.3) has the form

$$\theta^C = \theta + \Sigma_C^\dagger q + z, \quad z \in \ker(\Sigma_C),$$

because $q \in \mathcal{V}_C = \text{range}(\Sigma_C)$. The true target effect is

$$\psi_T = b_T + \ell_T^\top \theta + \ell_{U,T}^\top \gamma,$$

which follows by averaging (7.1) over the target covariate law. If $c_T \in \text{aff}(\mathcal{S})$, then $\ell_T \in \text{range}(\Sigma_C)$ by Lemma 1, so $\ell_T^\top z = 0$. Subtracting $\psi_T^C = b_T + \ell_T^\top \theta^C$ gives (7.5). $\square$

A.7 Proof of Theorem 2

Proof Subtract the affine combination of the source conditional treatment responses from the candidate response. The $g_\Delta(x)$ terms cancel because $\sum_s w_s = 1$, and the context terms cancel because $c^{(k)} = \sum_s w_s c_s$. The remaining term is

$$\left(u_k - \sum_s w_s u_s \right)^\top \gamma = d_{U,k}^\top \gamma,$$

which proves (7.7).

The candidate's observed contrast therefore adds exactly the equality $d_{U,k}^\top \gamma = \rho_k$ to the source moment and restriction system, which proves (7.10). The true parameter belongs to both sets by assumption, so the candidate set is a subset of the source set and both contain the truth. The infimum of the target functional can only increase and its supremum can only decrease after intersecting with this hyperplane, proving (7.11). If $d_{U,k} = 0$, the contrast is zero under the model and the added equality is redundant, so the sets and widths are equal. If the hyperplane cuts a target-relevant direction, one of the endpoint programs changes strictly. $\square$

A.8 Proof of Theorem 1

Proof By Proposition 4, $\mathcal{F}'$ is either $\mathcal{F}$ or $\mathcal{F}$ intersected with the two hyperplanes through β° , so $\mathcal{F}' \subseteq \mathcal{F}$ in both cases. The true vector β° satisfies the equality systems of both pools by (A.1) and lies in $\mathcal{R}$ by hypothesis, so $\beta^\circ \in \mathcal{F}'$ and both programs are feasible. By Proposition 4 the augmented-pool endpoints equal the base objective optimized over $\mathcal{F}'$, so shrinking the feasible set weakly raises the infimum and weakly lowers the supremum at every target, which is (6.1), with equality throughout in the rank-preserving case where $\mathcal{F}' = \mathcal{F}$. Since $\beta^\circ \in \mathcal{F}'$, the value $\psi_T = b_T + \ell_T^\top (\beta_1^\circ - \beta_0^\circ)$ lies between the augmented endpoints. $\square$

A.9 The dual price of the forecast

For completeness we record the sensitivity statement behind (5.3). Fix one polyhedral branch and a target, and consider the lower endpoint program of (5.2) as a function of the added right-hand side $y = (y_0, y_1)$,

$$v_-(y) = b_T + \min_{\beta} \left\{ \ell_T^\top (\beta_1 - \beta_0) : \Sigma_C \beta_a = d_a, u_k^\top \beta_a = y_a, a \in \{0, 1\}, G\beta \leq h \right\}.$$

This is a linear program whose right-hand side is affine in y , so v_- is a piecewise linear convex function of y on its feasibility domain, and standard linear-programming duality gives, at any y where the optimal basis is unique, $\partial v_-/\partial y_a = \lambda_{-,a}^u$, the optimal multiplier on the row $u_k^\top \beta_a = y_a$ [Bonnans and Shapiro, 2013, Chapter 4]. The upper endpoint is the corresponding concave value with multipliers $\lambda_{+,a}^u$. At a kink, the left and right slopes are the extreme multipliers, and the exact range of predicted intervals over $|y_a - \hat{y}_a| \leq B$ is obtained by re-solving the two programs at the corners, which is how the simulations and the case study evaluate forecast uncertainty. For a finite union of branches the value is the minimum or maximum of finitely many such functions and the same statements hold branchwise.